



Paralog editing tunes rice stomatal density to maintain photosynthesis and improve drought tolerance

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Abstract

Rice (*Oryza sativa*) is of paramount importance for global nutrition, supplying at least 20% of global calories. However, water scarcity and increased drought severity are anticipated to reduce rice yields globally. We explored stomatal developmental genetics as a mechanism for improving drought resilience in rice while maintaining yield under climate stress. CRISPR/Cas9-mediated knockouts of the positive regulator of stomatal development *STOMAGEN* and its paralog *EPIDERMAL PATTERNING FACTOR-LIKE10* (*EPFL10*) yielded lines with ~25% and 80% of wild-type stomatal density, respectively. *epfl10* lines with moderate reductions in stomatal density were able to conserve water to similar extents as *stomagen* lines but did not suffer from the concomitant reductions in stomatal conductance, carbon assimilation, or thermoregulation observed in *stomagen* knockouts. Moderate reductions in stomatal density achieved by editing *EPFL10* present a climate-adaptive approach for safeguarding yield in rice. Editing the paralog of *STOMAGEN* in other species may provide a means for tuning stomatal density in agriculturally important crops beyond rice.

Introduction

Prolonged periods of drought and increased desertification are anticipated to become more prevalent in the next century (IPCC 2019). Climate change modeling predicts increases in global temperatures by 2 to 4 °C by the end of the 21st century. Increased temperatures alone and in combination with limited water will negatively impact crop yields (Rang et al. 2011; Shah et al. 2011; Jagadish et al. 2015). The development of climate change adapted crops is essential to maintain crop yields in the face of rapid global population growth and worsening climates.

Rice (*Oryza sativa*) is 1 of the most widely consumed crops globally and trails only wheat and maize in area harvested (Food and Agriculture Organisation; FAO 2020). Originally domesticated in semiaquatic habitats, rice is especially

sensitive to drought relative to other C3 cereal crops (Lafitte et al. 2004; Bernier et al. 2008; Jägermeyr et al. 2021). Future water limitations may necessitate transitions of fully flooded paddy fields to water-saving production schemes (Jagadish et al. 2015). Rain-fed production, which comprises about 45% of total rice grown, is particularly susceptible to drought as a result of unpredictable precipitation (Khush 1997; Pandey et al. 2007; Tuong and Bouman 2009). Furthermore, most regions where irrigated rice is produced are currently experiencing or projected to experience water scarcity (Tuong and Bouman 2009). Thus, all rice, regardless of production methods, would benefit from improvements that maintain yields with less water.

Stomata are at the nexus of plants and the atmosphere. They facilitate gaseous exchanges of carbon dioxide, oxygen,

and water vapor. Most water loss in crops occurs via transpiration from the stomata (Buckley 2005). Breeding efforts have shown that drought adaptation in rice was facilitated in part by reductions of stomatal density (Kumar et al. 2021). Thus, opportunity exists, especially in nonadapted cultivars, to fine-tune stomatal density reductions.

The developmental biology of stomata has been studied extensively in myriad species (Liu et al. 2009; Endo and Torii 2019; Nunes et al. 2019; Harris et al. 2020). Despite similar sequences, EPIDERMAL PATTERNING FACTORS (EPFs) and EPF-LIKE proteins (EPFLs) mediate opposing downstream stomatal development responses. *EPF1* and *EPF2* function as negative regulators expressed in stomatal lineage cells. In contrast, *EPFL9*, also known as *STOMAGEN*, is a positive regulator of stomatal development that is dynamically expressed in the mesophyll (Hara et al. 2007, 2009; Hunt and Gray 2009; Hunt et al. 2010; Sugano et al. 2010; Shimada et al. 2011; Yin et al. 2017). These mobile peptides regulate cell fate transitions and cell divisions to ensure proper spacing and number of stomata (Hara et al. 2007, 2009; Hunt and Gray 2009; Sugano et al. 2010; Shimada et al. 2011; Lee et al. 2015).

EPFL9 is composed of 3 distinct regions: an N-terminal signal peptide region, a pro-peptide region, and a C-terminal cysteine-rich active peptide region (Sugano et al. 2010; Ohki et al. 2011). The full-length peptide is processed in vivo to yield a 45-residue C-terminal active peptide (Sugano et al. 2010; Lee et al. 2015). The active peptide encoded by *EPFL9* possesses the conserved cysteine residues of *EPF1* and *EPF2* and binds the same *ERECTA* (ER) family receptors and coreceptor TOO MANY MOUTHS (*TMM*) in *Arabidopsis* (*Arabidopsis thaliana*) (Sugano et al. 2010; Shimada et al. 2011; Lee et al. 2015; Hepworth et al. 2018). A knockout of *EPFL9* in rice using CRISPR/Cas9 and CRISPR/Cpf1 yielded an 8-fold reduction in abaxial stomatal density in the IR64 background (Yin et al. 2017). Likewise, a knockout of *EPFL9-1* in grape (*Vitis vinifera*) reduced stomatal density significantly (Clemens et al. 2022).

Interestingly, *EPFL9* appears to have undergone a duplication event in grasses (Hepworth et al. 2018). Heterologous expression of rice *STOMAGEN* and its duplicate, previously named *EPFL9-2*, in *A. thaliana* revealed a shared, though attenuated, function of *EPFL9-2* as a positive regulator of stomatal development when ectopically expressed (Lu et al. 2019). Ectopic expression of *Brachypodium distachyon* and wheat (*Triticum aestivum*) *STOMAGEN* and *STOMAGEN* paralogs in *A. thaliana* resulted in similar stomatal density increases (Jangra et al. 2021). In contrast, expression of negative regulators of stomatal development reduced stomatal density. Stomatal density reductions improved water use efficiency in *A. thaliana*, wheat, barley (*Hordeum vulgare*), and rice (Franks et al. 2015; Hepworth et al. 2015; Hughes et al. 2017; Caine et al. 2019; Dunn et al. 2019; Mohammed et al. 2019). However, all stomatal density reductions achieved by expressing negative regulators of stomatal development to any extent also reduced stomatal conductance and carbon

assimilation under physiologically relevant light conditions (Franks et al. 2015; Hughes et al. 2017; Dunn et al. 2019). For example, rice lines overexpressing *EPF1* to reduce stomatal densities exhibited lower stomatal conductance and carbon assimilation at all light conditions that exceeded 1,000 $\mu\text{mol photons/m}^2/\text{s}$ (Caine et al. 2019). These reductions may be detrimental as stomatal conductance has been shown to be essential for crop productivity (Fischer et al. 1998; Richards 2000; Ohsumi et al. 2007; Taylaran et al. 2011; Roche 2015; Kusumi et al. 2017; Faralli et al. 2019; Zhang et al. 2021). In rice specifically, higher stomatal conductance has been associated with greater rates of leaf photosynthesis (Ohsumi et al. 2007; Kusumi et al. 2017; Zhang et al. 2021). Also of note, all previously tested strategies relied on the use of constitutive transgenic expression, lacking the tissue specificity and gene dosage that may regulate stomatal density phenotypes in vivo.

Gene editing of rice *STOMAGEN* and its duplicate were used as an alternative, nontransgenic approach to tune stomatal density. Gene duplication events can provide genetic material for functional novelty (Conant and Wolfe 2008), but may also disrupt optimal levels of gene expression, requiring substantial evolutionary time to stabilize (Qian et al. 2010). Gene editing of paralogs can provide a potent and straightforward mechanism to generate desirable variation in a wide array of traits. For example, knockout alleles of individual genes in a set of paralogs underlying disease resistance and yield in tomato and maize, respectively, have demonstrated the utility of paralog gene editing for generating desirable trait variation (de Toledo Thomazella et al. 2021; Liu et al. 2021).

Here, we report characterization of the rice *STOMAGEN* paralog *EPFL9-2*, subsequently referred to as *EPFL10*, and its relationship to stomatal development. Furthermore, we explore the effect of the reductions in stomatal density in *stomagen* and *epfl10* on stomatal conductance, carbon assimilation, thermal regulation, water conservation, and yield in varying water regimes. We describe the use of paralog editing for achieving desirable variation in traits of interest and the implications of this work on future gene-editing strategies for improved water-use efficiency.

Results

Evolution and regulation of *STOMAGEN* and *EPFL10* suggest functional differences of paralogs

Previous work has suggested that the Os*STOMAGEN* paralog Os*EPFL10* (LOC_Os08g41360) may function as a weak, positive regulator of stomatal development (Lu et al. 2019; Jangra et al. 2021), having arisen from a putative duplication event in the most recent common ancestor of the Poaceae species (Hepworth et al. 2018). We examined evolutionary divergence of *STOMAGEN* and *EPFL10* that may explain such functional differences. The 2 paralogs have accumulated divergent point mutations. In rice, homology cannot be

detected between *STOMAGEN* and *EPFL10* across the N-terminal signal peptides and pro-peptide domain. The functional 45 C-terminal sequences show 73% sequence identity with 12 nonsynonymous substitutions (Fig. 1C). These positions were mapped onto the existing complex structure of *TOO MANY MOUTHS* (TMM), *ERECTA-like 1*, and *EPF1* (Fig. 1D), assuming the binding of *STOMAGEN* and *EPFL10* may be similar to *EPF1*. Many positions highlighted with nonsynonymous substitutions were in proximity to the receptors, possibly suggesting that the paralogs may have different binding affinities to the receptors.

The phylogenetic tree also indicated that the 2 orthologous groups may have experienced differential selective pressure (Fig. 1A). To gain better insights, we calculated sequence variations of the *STOMAGEN* and *EPFL10* orthologs with greater numbers of sequences collected from the *Oryza* species and the Poaceae family members (Fig. 1C and Supplemental Table S1). Consistent with the phylogeny, both species- and family-level sequence diversities were higher in *EPFL10* orthologs, indicating relatively rapid divergence across the species. On the contrary, *STOMAGEN* orthologs have maintained a high level of sequence conservation across the species and family, suggesting their sequence evolution is strictly constrained. Furthermore, regulatory variations existed between *STOMAGEN* and *EPFL10*. Repressive CHG and CHH methylation marks vicinal to the *EPFL10* coding region suggested regulatory restriction, while this was not observed for *STOMAGEN*. Concordantly, we observed that *STOMAGEN* mRNA abundance also greatly exceeded *EPFL10* expression in leaf base tissues where stomatal development occurs and *STOMAGEN* and *EPFL10* expression is greatest (Fig. 1B). There is minimal expression of either transcript in adult leaf where stomatal complexes are fully matured (Fig. 1B). Collectively, the sequence evolution and regulation of *STOMAGEN* and *EPFL10* suggested possibly differential evolution and usage of the paralogs. This suggested that *EPFL10* may be a better candidate to fine-tune stomatal development without abolishing the role of *STOMAGEN*, the primary positive regulator of stomatal development.

Stomatal density and morphology are altered in knockout lines

To clarify the functional role of *OsSTOMAGEN* and *OsEPFL10* in regulating stomatal density in vivo, single- and double-mutant lines were generated in rice. CRISPR/Cas9-mediated knockout of *OsSTOMAGEN* and *OsEPFL10* was achieved by targeting guides to the first exon of each gene to disrupt the open reading frame (Fig. 1, E and F). Two unique homozygous knockout alleles were generated in *OsEPFL10* and in *OsSTOMAGEN* in the T_0 generation using a single guide sequence adjacent to a PAM motif (Fig. 1, E and F).

epfl10 exhibited reductions in stomatal densities which represented 80% of wild-type densities, whereas *STOMAGEN* possessed only 25% of wild-type densities in the fifth fully expanded adult leaf (Figs. 2A and S1). Similar proportions of

stomatal density reductions in mutants were measured in flag leaves and their adaxial leaf surfaces (Fig. 2, B and C). Stomatal length was measured to determine if there was a relationship between stomatal density reductions and size increases in the Nipponbare background. The guard cell length of stomata in *stomagen* was longer relative to *epfl10* and wild type (Fig. 2D). Representative confocal microscopy captured stomatal density relationships in a single biological replicate of each genotype (Fig. 2E). The stomatal density of *stomagen epfl10* double mutants could not be distinguished from *stomagen* and was thus excluded from future physiological analyses (Supplemental Fig. S2). Collectively, our data suggest that *OsEPFL10* plays a functional role in rice stomatal formation, but the loss of *OsSTOMAGEN* is epistatic to *EPFL10*-mediated stomatal development.

Stomatal density reductions are concomitant with gas exchange reductions in *stomagen* but not *epfl10*

The assayed genotypes revealed anatomical level differences in stomatal densities and length. To determine if these developmental differences corresponded to alterations in carbon assimilation (A_n) and stomatal conductance (g_s), the efficiency of gas exchange was measured across genotypes in response to increasing incident light.

At ambient CO_2 (400 ppm), *epfl10* maintained wild-type levels of A_n and g_s whereas *stomagen* exhibited reduced gas exchange capacity in light response curves relative to *epfl10* and wild type at all light intensities greater than 1,000 $\mu\text{mol photons/m}^2/\text{s}$ (Fig. 3, A and B). In an independent cohort of plants assayed at saturating light (1,000 $\mu\text{mol photons/m}^2/\text{s}$), *stomagen* steady-state A_n and g_s were lower relative to wild type and *epfl10* grown in growth chambers (Supplemental Figs. S3, S4 and S5). Similar reductions of A_n and g_s in *stomagen* lines were recapitulated in greenhouse measurements at 1,000 $\mu\text{mol photons/m}^2/\text{s}$ (Supplemental Figs. S6 and S7). However, under vegetative drought, *stomagen* did not exhibit lowered levels of carbon assimilation at 1,000 $\mu\text{mol photons/m}^2/\text{s}$ despite reduced levels of stomatal conductance (Supplemental Figs. S6 and S7).

Measurements of stomatal apertures on the abaxial side of flag leaves indicated that *epfl10* lines maintained a larger stomatal aperture than the wild type, and *stomagen* lines exhibited an even greater aperture (Fig. 3, C and D). Thus, *EPFL10* and *STOMAGEN* maintained even greater levels of stomatal conductance per individual stoma mediated by a larger aperture in well-watered conditions (Fig. 3E). Increased stomatal apertures at reductions in stomatal density of 25% but not 80% were able to sustain wild-type levels of steady-state A_n and g_s .

Decreasing stomatal density increases ratio of theoretical stomatal conductance to operating stomatal conductance

Despite similar operating stomatal conductance, *epfl10* and wild type significantly differ in their theoretical maximum stomatal conductance (Fig. 3F). This is in contrast to

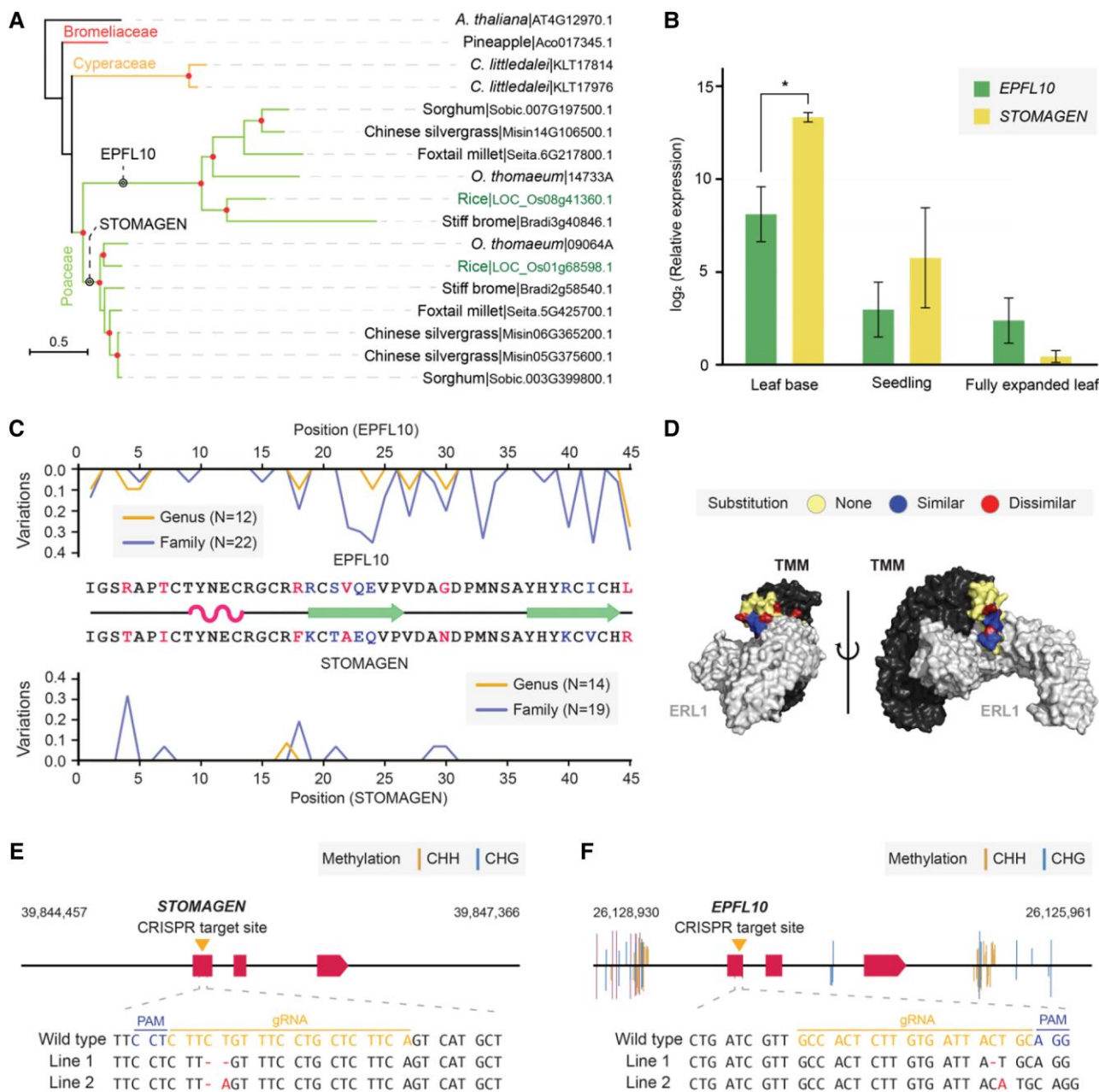


Figure 1. Evolution and regulation of *STOMAGEN* and *EPFL10*. **A**) Gene trees of the *STOMAGEN* family members in Poales. *Arabidopsis thaliana* was used as an outgroup. Full-length *STOMAGEN* and *EPFL10* orthologs were aligned with MUSCLE v3.8.1551. (Edgar 2004). The positions with 40% or more gaps were removed. The gene tree was inferred with the filtered alignment with fasttree v2.1.10 (-gamma -slow -wag), and *Arabidopsis* *STOMAGEN* was used as an outgroup to root the tree. The scale bar indicates the branch length for 0.50 substitutions per site. **B**) RT-qPCR determined expression levels of *EPFL10* and *STOMAGEN* in fully expanded leaf, leaf base, and seedling. Bars represent means and error bars represent so from the mean. The asterisk indicates a significant difference between the means ($P < 0.05$). **C**) Genus and family-level sequence variations of *STOMAGEN* and *EPFL10* orthologs. The sequence variations represent normalized Shannon's entropy, with 0 being no variations and 1 being complete variations. The secondary structure annotation originates from the experimentally determined structure of *STOMAGEN* (PDB: 2LIY) (Ohki et al. 2011). The colored residues highlight variable positions between rice *STOMAGEN* and *EPFL10*. Substitutions to similar and dissimilar amino acids based on BLOSUM62 are indicated with blue and red, respectively. Alpha helix is represented by pink and beta sheets by green. **D**) The variable positions between *STOMAGEN* and *EPFL10* mapped to the experimentally determined complex structure of EPF1, ERL1 (in gray), and TMM (in black) (PDB: 5XJO) (Lin et al. 2017). Peptide sequence variations between *EPFL10* and *STOMAGEN* with similar or dissimilar substitutions are indicated in blue and red, respectively. Complex is visualized from 2 orientations. **E** and **F**) The gene models of **E**) *STOMAGEN* and **F**) *EPFL10* with the location of the CRISPR/Cas9 target sites indicated by the yellow triangle. Repressive CHH and CHG methylation marks are indicated for 1.5 kb up- and downstream of the coding regions. Methylation data were collected from 3-wk-old leaf tissues. The 2 unique homozygous edits generated in *STOMAGEN* and *EPFL10* by CRISPR/Cas9 are shown in red with the PAM annotated in blue and guide sequence in yellow.

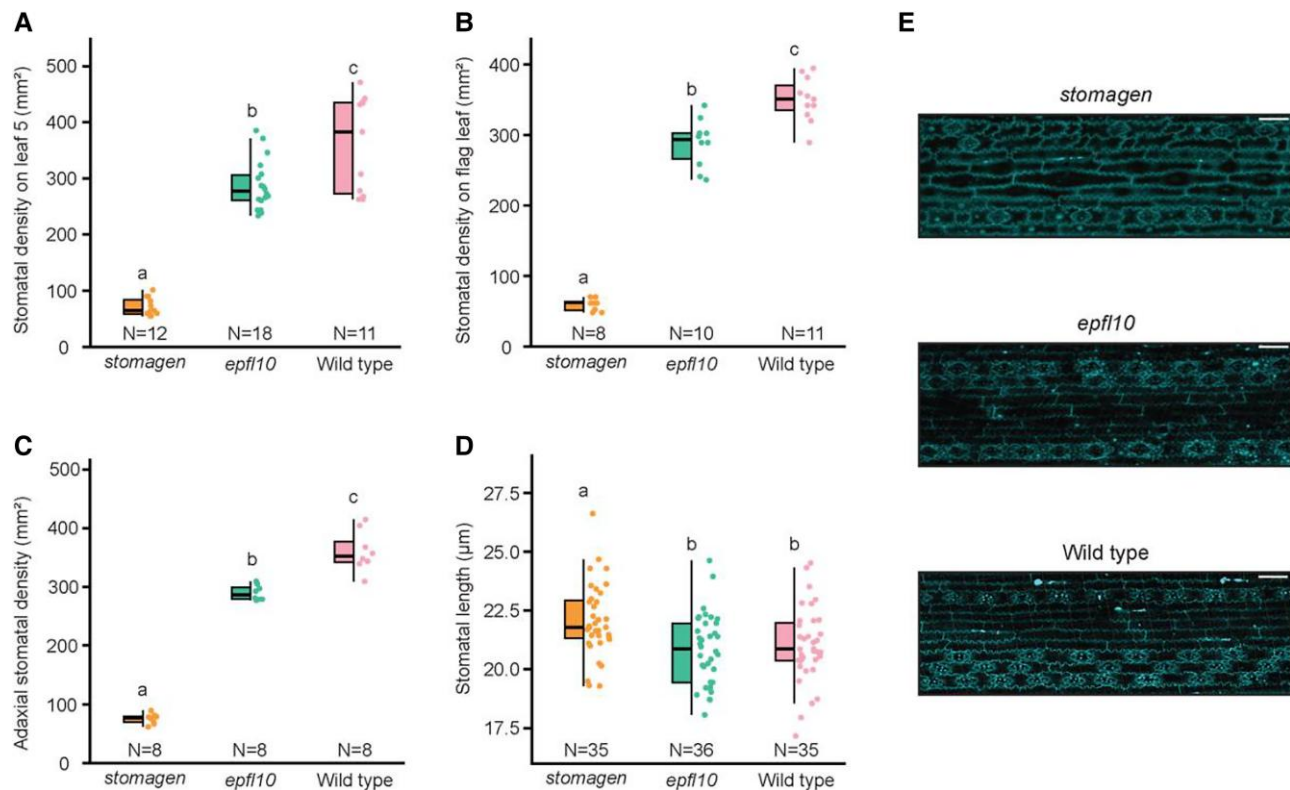


Figure 2. EPFL10 is a weak positive regulator of stomatal development in Nipponbare (*Oryza sativa* spp. japonica). **A to C)** Stomatal density of *epfl10*, *stomagen*, and wild type. **A)** Stomatal density of the fifth fully expanded true leaf. Stomatal densities of 21-d-old plants were measured. **B)** Stomatal density of the flag leaf on the primary tiller during grain filling. Flag leaves of 55-d-old plants were measured. **C)** Adaxial stomatal densities of the fifth fully expanded true leaf on 21-d-old plants. **D)** Stomatal length of *epfl10*, *stomagen*, and wild type. **E)** Representative confocal images of *epfl10*, *stomagen*, and wild type. Images were taken of 55-d-old leaves stained with propidium iodide. **A to D)** Measurements were taken on pooled alleles of *epfl10* and *stomagen*. Graphs are box-and-whisker plots where the center horizontal indicates the median; upper and lower edges of the box are the upper and lower quartiles; and whiskers extend to the maximum and minimum values within 1.5 interquartile ranges. Letters indicate a significant difference between means ($P < 0.05$, 1-way ANOVA Tukey HSD post hoc test).

stomagen which had both significantly lower operating and anatomical stomatal conductance. Thus, reductions in stomatal density result in a stomatal conductance that operates at a greater proportion of its theoretical maximum for both *stomagen* and *epfl10*, though the latter is able to compensate to sustain WILD-TYPE gas exchange (Fig. 2, A and B).

EPFL10 maintains wild-type thermoregulation and yield while increasing water conservation

Measurements of A_n and g_s and $g_{s\max}$ revealed differences in the regulation of gas exchange among wild type, *epfl10*, and *stomagen*. To further resolve the lifetime implications of these edits, we assessed differences in thermoregulatory, yield, and water conservation capacities.

Thermal imaging was used to assess evaporative cooling in lines with altered stomata. In well-watered conditions, *stomagen* lines were warmer on average than wild type, whereas *epfl10* leaf temperatures were wild type like or cooler (Fig. 4A). No difference in leaf temperature was detected across genotypes during vegetative or reproductive drought. To test the impact of reduced stomatal densities on water conservation, daily water loss was measured over the course

of a week beginning with 70 d after germination. Despite differences in thermoregulation, *stomagen* and *epfl10* both conserved greater volumes of water in a week by 48 mL and 83 mL, respectively (Fig. 4D).

epfl10 exhibited a range of phenotypes that suggested an increased fitness under drought via moderate stomatal density reductions. This phenotype contrasted with *stomagen*, which may be less optimal for high-yielding production due to large reductions in A_n and g_s . To assess the impacts of stomatal modifications on potential crop performance, yield trials were conducted using 3 watering regimes in the greenhouse. However, in well-watered, vegetative drought, and 4-d reproductive drought conditions, there was no significant difference in grain yield or aboveground biomass among genotypes (Fig. 4, B and C).

Duplicates of STOMAGEN orthologs in other species may also be targets for tuning stomatal density

The potential of paralog editing for tuning stomatal density, as demonstrated in rice *epfl10* lines, can be extended to other crop species. Further phylogenetic investigation of the duplication of STOMAGEN among angiosperms

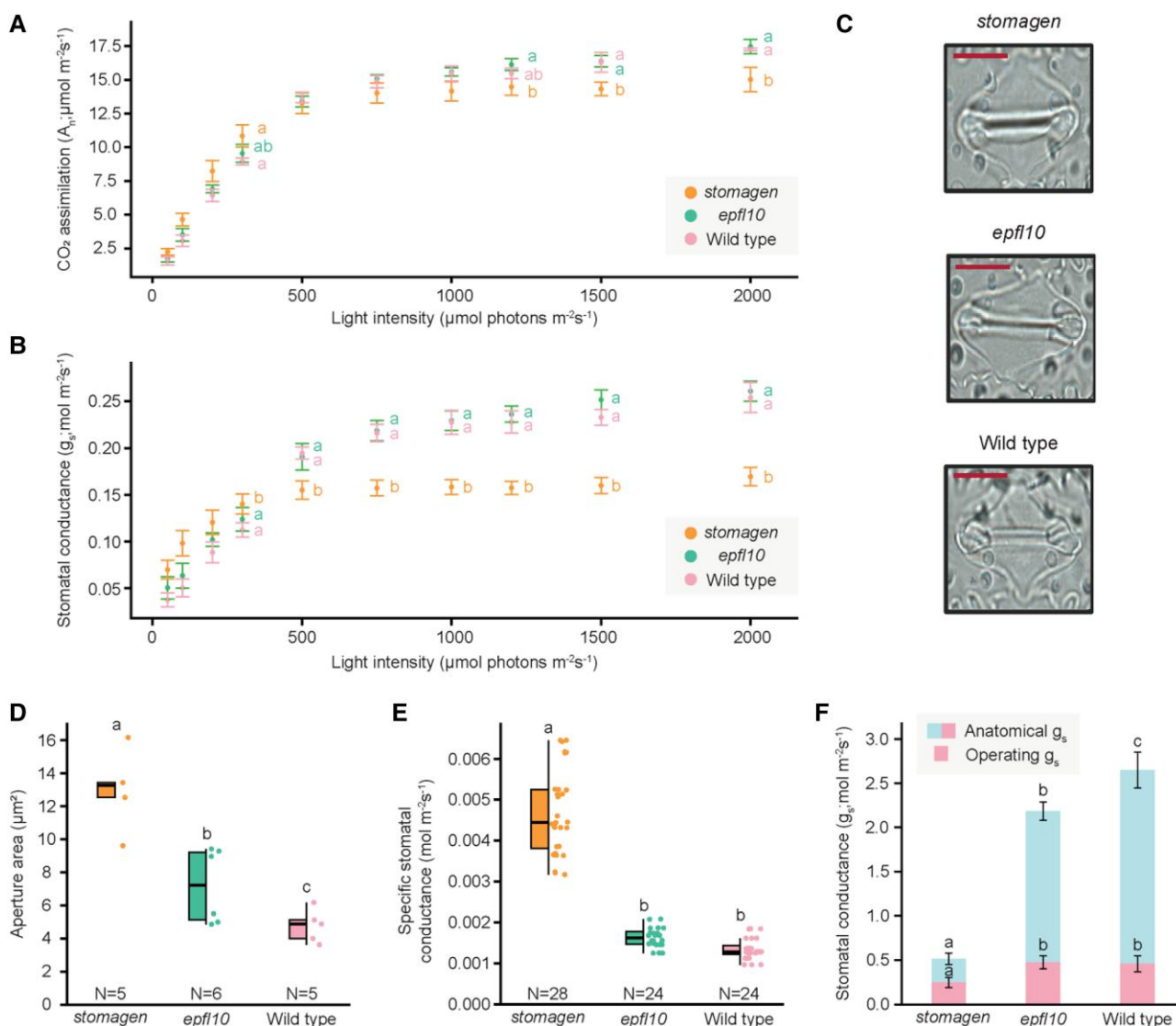


Figure 3. Gas exchange measurements and stomatal pore area measurements in reduced stomatal density backgrounds in Nipponbare (*Oryza sativa* spp. *japonica*). **A** and **B**) Carbon assimilation **A**) and stomatal conductance **B**) measurements of *STOMAGEN*, *EPFL10*, and wild type across a range of light intensities: 100, 200, 300, 500, 650, 1,000, 1,200, 1,500, and 2,000 $\mu\text{mol photons/m}^2/\text{s}$ in 32-d-old plants grown in the greenhouse. **C**) Representative images of stomatal pore size variations for *EPFL10*, *STOMAGEN*, and wild type. **D**) Stomatal aperture of *EPFL10*, *STOMAGEN*, and wild type. **E**) Specific stomatal conductance of *EPFL10*, *STOMAGEN*, and wild type. Specific conductance values were calculated by dividing stomatal conductance by the average stomatal density of the probe area of the respective genotype. **F**) Anatomical $g_{s\text{max}}$ was calculated for each genotype. Operating g_s at 1,500 $\mu\text{mol photons/m}^2/\text{s}$ is reported in pink. The percentage of operating to anatomical g_s is reported above the pink bar for each genotype. **B** and **D**) Dots represent means and error bars are SD from the mean. **C**, **E**, and **F**) In the box-and-whisker plots, the center horizontal indicates the median; upper and lower edges of the box are the upper and lower quartiles; and whiskers extend to the maximum and minimum values within 1.5 interquartile ranges. Letters indicate a significant difference between means ($P < 0.05$, 1-way ANOVA Tukey HSD post hoc test).

revealed an additional putative family-level *STOMAGEN* duplication in the Asteraceae (Fig. 5A). Species-level duplications were noted in carrot (*Daucus carota*), date palm (*Phoenix dactylifera*), Balbis banana (*Musa balbisiana*), and *Carex littledalei* based on the species included in this study.

Similar to rice *STOMAGEN* and *EPFL10*, the paralogs of *STOMAGEN* in carrot and sunflower are differentially expressed in tissues most relevant to gas exchange suggesting regulatory divergence after the duplication event

(Fig. 5, B and C). Unlike the rice *STOMAGEN* duplicates, both sunflower and carrot paralogs exhibit less relative sequence variation (Fig. 5, D and E). Regardless of functional divergence, duplicated *STOMAGEN* copies may provide genetic material for optimization of expression in organisms where multiple copies exist.

Discussion

Drought is the most severe and widespread environmental stressor in South and Southeast Asia (Climate change—ready

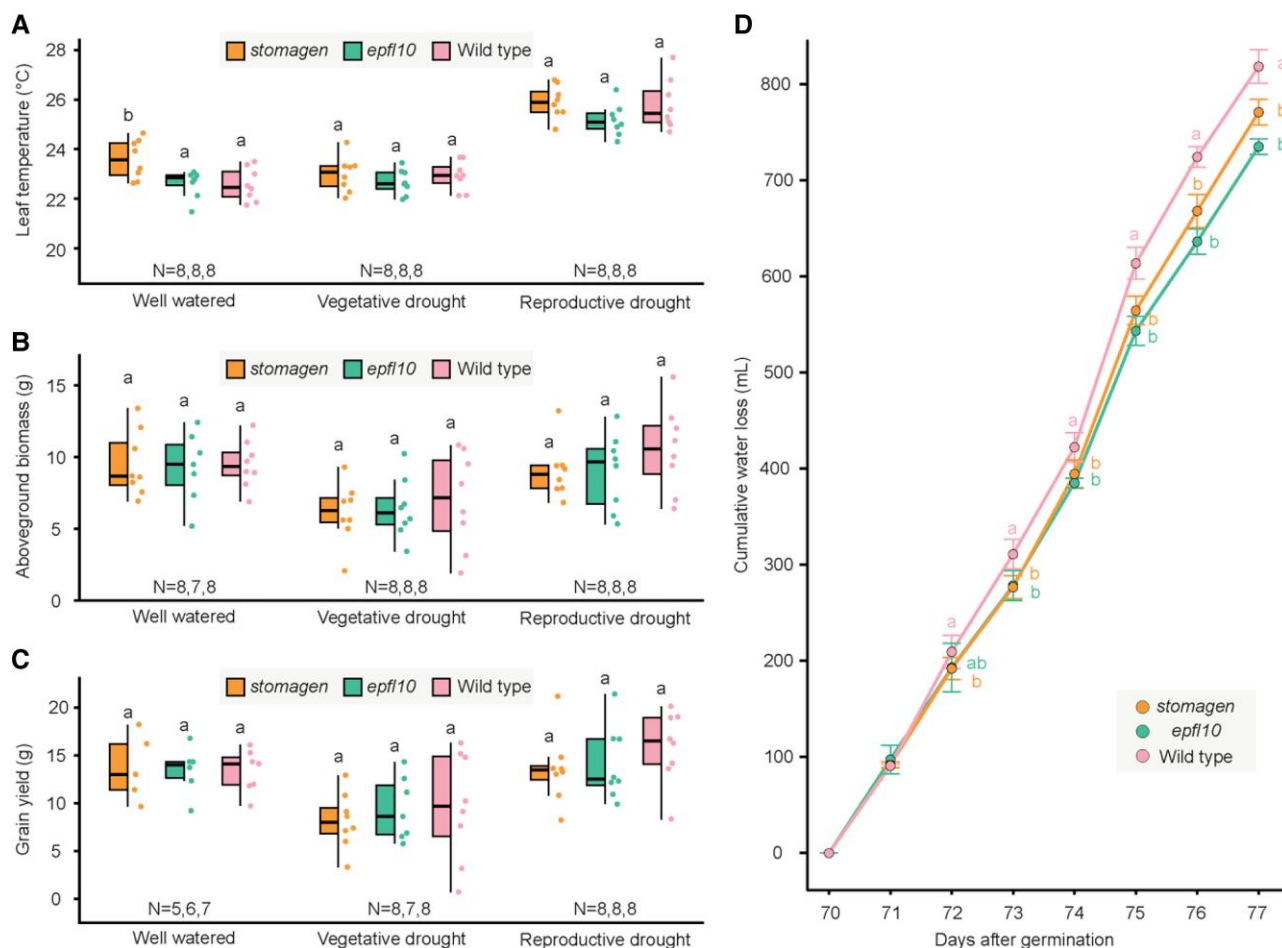


Figure 4. Stomatal density reductions influence thermoregulation and water conservation but not greenhouse yield in Nipponbare (*Oryza sativa* spp. *japonica*). **A)** Leaf temperatures of *STOMAGEN*, *EPFL10*, and wild type in well-watered, vegetative drought, and reproductive drought. **B)** Aboveground biomass of *STOMAGEN*, *EPFL10*, and wild type in well-watered, vegetative drought, and reproductive drought. **C)** Grain yield of *STOMAGEN*, *EPFL10*, and wild type in well-watered, vegetative drought, and reproductive drought. **A to C)** Graphs are box-and-whisker plots where the center horizontal indicates the median; upper and lower edges of the box are the upper and lower quartiles; and whiskers extend to the maximum and minimum values within 1.5 interquartile ranges. **D)** Cumulative water loss of *STOMAGEN*, *EPFL10*, and wild type from Days 70 to 77 after germination. Pots remained well watered during the course of this experiment. Flats were replenished after daily water loss recordings. The dots represent means and error bars are *se* from the mean. Letters indicate a significant difference between means ($P < 0.05$, 1-way ANOVA Tukey HSD post hoc test).

rice | International Rice Research Institute no date). Application of gene editing in crops for climate change could serve as a potent mechanism for realizing actual technology transfer to growers. Few gene-editing applications for addressing abiotic stressors have been reported to date (Karavolias et al. 2021). Preceding transgenic manipulations of epidermal patterning factors generated rice with improved water conservation and maintenance of yields in greenhouse conditions; however, these lines exhibited reductions in A_n and g_s (Hughes et al. 2017; Caine et al. 2019; Dunn et al. 2019). Exploration of the rice *STOMAGEN* paralog, *OsEPFL10*, was undertaken as a means of generating novel, transgene-free variation in rice stomatal density. The evolutionary and regulatory features of *EPFL10* suggested it as a promising target for engineering drought resilience.

Variation in stomatal density introduced by single-gene knockouts of *OsEPFL10* and *OsSTOMAGEN* provided a basis for further physiological exploration. A double knockout of *OsEPFL10* and *OsSTOMAGEN* was also generated. *stomagen epfl10* double mutants phenocopy *stomagen* in stomatal density, suggesting that loss of *STOMAGEN* may be epistatic to *EPFL10*. However, the stomatal density reductions of *stomagen* lines are so severe that *STOMAGEN* knockout alleles may mask the subtle contribution of *EPFL10* towards stomatal density. Further investigations of *EPFL10* and *STOMAGEN* interactions are needed to clarify the relationship of these genes.

Stomatal conductance and photosynthesis of *stomagen* but not *epfl10* were lower at all light intensities greater than 1,000 $\mu\text{mol photons/m}^2/\text{s}$ relative to wild type. In field

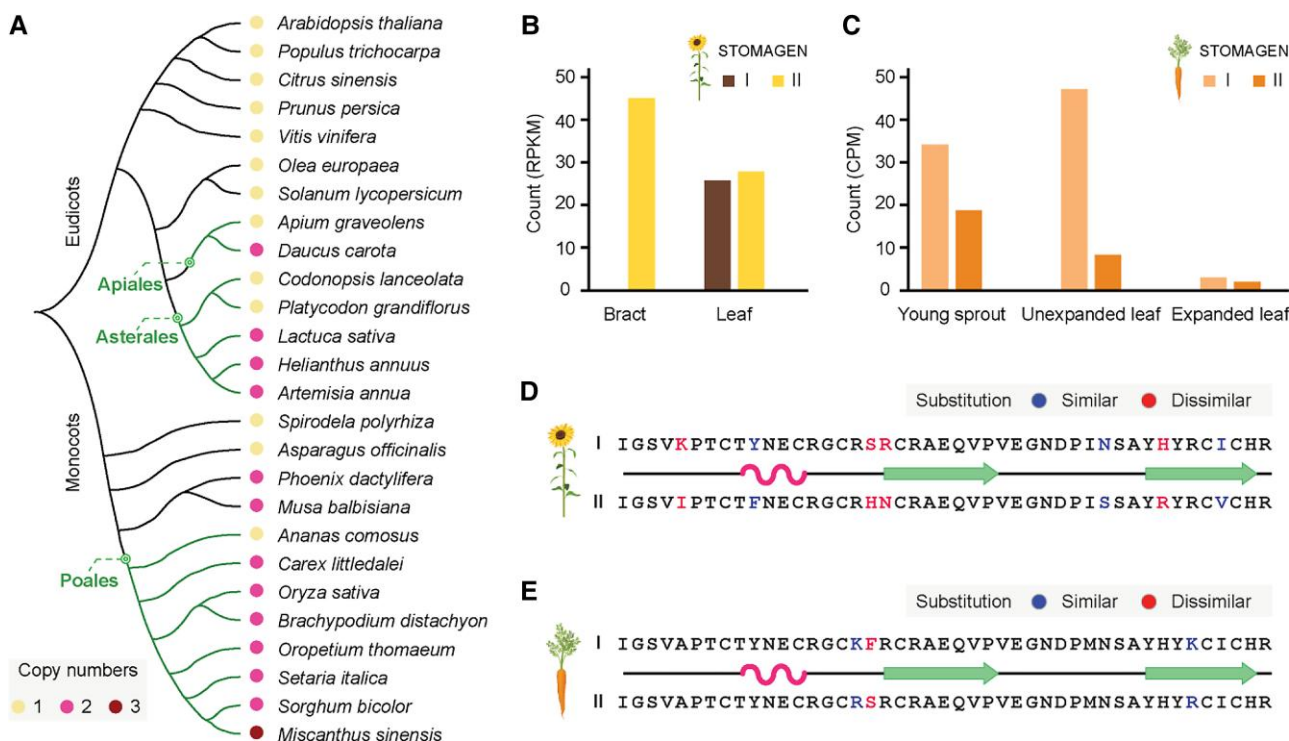


Figure 5. Multiple STOMAGEN duplication events in angiosperms provide new editing targets. **A**) A dendrogram of angiosperm species and copy number variations of STOMAGEN family members. STOMAGEN copy number indicated by colors in legend. **B** and **C**) Expression patterns of the paralogous STOMAGENs in **B**) sunflower (*Helianthus annuus*) and **C**) carrot (*Daucus carota*) in tissues assayed most relevant to gas exchange. **D** and **E**) The C-terminal-active STOMAGEN sequences with nonsynonymous substitutions highlighted for the paralogs in sunflower **D**) and carrot **E**). Substitutions to similar and dissimilar amino acids based on BLOSUM62 are indicated with blue and red. The secondary structure annotation originates from the experimentally determined solved structure of STOMAGEN (PDB: 2LIY). Alpha helix is represented by pink and beta sheets by green.

conditions, photosynthetic flux densities regularly exceed $1,000 \mu\text{mol photons/m}^2/\text{s}$ at midday, coinciding with the hottest parts of the day (Slattery et al. 2018). Sustaining photosynthesis may work synergistically with optimizing thermoregulation as *stomagen* exhibits reduced gas exchange simultaneous with warmer temperatures in well-watered conditions whereas *epfl10* maintained wild-type levels of thermoregulations (Figs. 3, A and B and 4A). In accordance with anticipated global warming caused by climate change, maintained or improved thermoregulation will be vital for crop agronomic performance (Jagadish et al. 2015).

Larger stomatal apertures were measured in *stomagen* and *epfl10* lines relative to wild type (Fig. 3D). Pore area adjustments in *stomagen* lines were unable to physiologically compensate for large reductions in stomatal densities, unlike *epfl10* lines, which maintained wild-type levels of A_n and g_s (Fig. 3, A and B). The theoretical maximum stomatal conductance of grass stomata greatly exceeds the measured stomatal conductance (Caine et al. 2019; Faralli et al. 2019). *stomagen* maintained high levels of specific stomatal conductance concurrent albeit with low overall A_n and g_s (Fig. 3, A, B, and E). High levels of specific stomatal conductance in *stomagen* reveal that engineering greater stomatal conductance in a reduced density background may be a

promising mechanism for maintaining photosynthetic capacities simultaneous with water-use efficiency (Buckley et al. 2019; Faralli et al. 2019).

No detectable differences were found in greenhouse grain or biomass yield among the reported genotypes under periods of vegetative or reproductive drought. However, there is a reasonable basis to prioritize maintenance of wild-type gas exchange for maximizing yield. Other literature suggests that high levels of leaf photosynthesis and stomatal conductance are linked to higher yields among C3 crops (Richards 2000; Ohsumi et al. 2007; Taylaran et al. 2011; Roche 2015; Kusumi et al. 2017; Zhang et al. 2021), and it is possible that more severe drought stress or in-field validation may resolve differences between genotypes.

Independent of biomass and yield, *epfl10* provides a potent demonstration of the capacity to reduce stomatal densities and increase soil water conservation without concomitant reductions in gas exchange essential for optimal crop performance at steady-state conditions. Stomatal conductance measurements are an instantaneous measurement of plants taken at steady state. However, dynamic environmental conditions exist in greenhouse conditions where water conservation measurements were made. Periods in this dynamic range where stomata may have been closed, such as at nighttime,

or less open, like periods of cloud cover, may have contributed the capacity of *epfl10* to conserve more water than WILD TYPE despite statistically similar stomatal conductance. Transpiration via closed stomata or less open stomata is a nonnegligible mechanism of water loss that is frequently overlooked and that may, in part, explain water loss phenotypes (Caird et al. 2007). Furthermore, stomatal density alone does not account for gas exchange entirely. Stomatal kinetics in dynamic environments has been shown to drive differences in conductance (Lawson and Vialet-Chabrand 2019; Eyland et al. 2021).

It is noteworthy that *stomagen* lines conserved water to a similar extent as *epfl10* despite having much greater reductions in stomatal density. Larger stomata typically exhibit slower dynamic responses (Drake et al. 2013). Slower rates of environmental response in fluctuating light or environmental conditions may account for a nonlinear relationship between stomatal density and water conservation. Our data indicate that fewer stomata with larger apertures offer comparable water conservation properties as lines possessing greater numbers of stomata with smaller apertures. It is still unclear how more severe and field-relevant drought stresses may affect water loss and assimilation when specific stomatal conductance is limiting.

Knockout of the *OsSTOMAGEN* paralog *OsEPFL10* builds on other work that has shown that null alleles of paralogs can achieve desirable phenotypic outcomes (de Toledo Thomazella et al. 2021; Liu et al. 2021). Rice *EPFL10* had reduced expression levels in relevant tissues and considerable peptide-level differences from *STOMAGEN*, reinforcing *ex situ* evidence that *OsEPFL10* may have evolved a weaker overall function in stomatal development (Fig. 1). A similar exploration of *STOMAGEN* duplicates in other organisms was used to identify other candidate species for paralog editing. *STOMAGEN* copy II in sunflower and *STOMAGEN* copy I in carrot also each exhibit lower expression that may substantiate these genes as targets for moderate reductions in stomatal density similar to *EPFL10* in rice. A previous report identified 2 copies of *STOMAGEN* in a collection of grape genotypes using high-quality published genomes and verifying by PCR. Advances in sequencing such as genome and pangenome availability may reveal other putative duplications at the species or cultivar level. Gene editing of paralogs can provide a convenient gene-editing target for quantitative variation in traits like stomatal density.

Peptide variation of the *STOMAGEN* duplicate, *EPFL10*, in rice exceeded that of carrot and sunflower (Fig. 5). It is enticing to consider a scenario where the substantial sequence variation underlying *OsEPFL10* at the family and genus level may evidence selection against functional redundancy of *STOMAGEN* (Fig. 1C). Thus, both decreases in expression, observed in all duplicates, and substantial peptide variation, observed in rice *EPFL10*, may underlie functional natural variation that knocked down stomatal density to restore gene dosage imbalances incurred by stochastic *STOMAGEN* duplication. Further functional characterization of diverse

STOMAGEN duplicates is necessary to resolve their role and costs to plant fitness. Paralog editing of stomatal density and other traits may be a promising avenue to accelerate the production of climate-adapted varieties without the constraints of transgenic germplasm.

To this aim, we report here modest stomatal density reductions in *epfl10* mutants with no associated decreases in stomatal conductance or carbon assimilation. *epfl10* lines maintained wild-type physiological capacities of stomatal conductance, carbon assimilation, thermoregulation, and yield while also conserving more water than wild type. These attributes could contribute to improved climate resilience in current and future conditions where water is limiting, and temperatures are increased. Field-based investigations of *epfl10* and *stomagen* will further resolve the agronomic utility of these edited rice lines.

Materials and methods

Plant growth conditions

Rice cultivar Nipponbare (*Oryza sativa* ssp. *japonica*) seeds were germinated and grown for 8 d in a petri dish with 20 mL of water in a Conviron growth chamber at 28 °C for day-length periods of 16 h in 100 $\mu\text{mol photons/m}^2/\text{s}$ of light and 80% relative humidity. Seedlings were transferred to a soil mixture comprised of equal parts turf (https://www.turf.com/products/infield-conditioners/mvp) and sunshine mix #4 (http://www.sungro.com/professional-products/fafard/).

Germinated seedlings used for stomatal phenotyping and growth chamber physiological assays were transferred to 10-cm, 0.75-L McConkey tech square pots and placed in growth chambers at 28 °C for day-length periods of 16 h in 400 $\mu\text{mol photons/m}^2/\text{s}$ of light and 80% relative humidity.

Plants designated for yield trials, greenhouse physiological assays, and stomatal aperture measurements were moved to the greenhouse with temperature set points of 27 °C day/22 °C night at ambient light conditions in February 2020 with day lengths of 12 h in 15.2-cm, 1.8-L pots. All plants were fertilized with 125 mL of 1% w/v iron solution 1-wk post-transplant. One liter of 5% w/v JR Peter's Blue 20-20-20 fertilizer (https://www.jrpeters.com/) was added to each flat containing 6 plants at 3 and 11 wk postgermination.

Yield and water regimes

Adapted from Caine et al. (2019), grain and biomass yield in 3 watering regimes were tested: well-watered, vegetative drought, and reproductive drought. Well-watered flats were kept flooded for the entirety of the growth period. Vegetative drought was imposed by removing all water from flats containing pots for 7 d starting on Day 28 after germination and for 9 d at Day 56 after germination. In reproductive drought, water was removed from flats for 4 d at Day 98 when panicles were undergoing grain filling. All grain and aboveground biomass from well-watered, vegetative, and reproductive drought plants were harvested after

167 d, 177 d, and 181 d, respectively. Biomass measurements were completed on samples dried at 60 °C for 3 d prior to weighing.

Generation of edited lines

Guides for targeting *EPFL10* and *STOMAGEN* were selected to minimize off-target effects and maximize on-target efficiency in the first exon of the coding region. Guide sequences were selected using CRISPR-P 2.0 (<http://crispr.hzau.edu.cn/CRISPR2/>). Forward and reverse strand guide sequences with relevant sticky ends amenable for Golden Gate cloning were ordered from Integrated DNA Technology (IDTdna.com). Equal volumes of 10 mM primers were annealed at room temperature. Golden Gate cloning was used to insert guides into the PeGM entry clone containing the tracrRNA and U3 promoter. LR clone reactions were used to insert the entry clone into the destination vectors for biolistic transformation and *Agrobacterium* (*Agrobacterium tumefaciens*)-mediated transformation. *epfl10* lines were produced via *Agrobacterium*-mediated transformation and *stomagen* lines via biolistic transformation. The OsEPFL10 editing vector and the OsSTOMAGEN editing vector maps are available as Supplemental Figs. S8 and S9, respectively, and the vector sequences are available in Supplemental Files S1 and S2.

Plant material and culture of explants

Mature seeds of rice (*O. sativa* L. *japonica* cv. Nipponbare) were de-hulled and surface-sterilized for 20 min in 20% (v/v) commercial bleach (5.25% sodium hypochlorite v/v) and 1% of Tween 20 v/v. Three washes in sterile water were used to remove residual bleach from seeds. De-hulled seeds were placed on callus induction medium (CIM) (N6 salts and vitamins (Chu et al. 1975), 30 g/L maltose, 0.1 g/L myo-inositol, 0.3 g/L casein enzymatic hydrolysate, 0.5 g/L L-proline, 0.5 g/L L-glutamine, 2.5 mg/L 2,4-D, 0.2 mg/L BAP, 5 mM CuSO₄, and 3.5 g/L Phytigel, pH 5.8) and incubated in the dark at 28 °C to initiate callus induction. Six- to 8-wk-old embryogenic calli were used as targets for transformation.

Agrobacterium-mediated transformation

Embryogenic calli were dried for 30 min prior to incubation with an *A. tumefaciens* EHA105 suspension (OD_{600nm} = 0.1) carrying the binary vector for editing rice *EPFL10*. After a 30-min incubation, the *Agrobacterium* suspension was removed. Calli were then placed on sterile filter paper, transferred to cocultivation medium (N6 salts and vitamins, 30 g/L maltose, 10 g/L glucose, 0.1 g/L myo-inositol, 0.3 g/L casein enzymatic hydrolysate, 0.5 g/L L-proline, 0.5 g/L L-glutamine, 2 mg/L 2,4-D, 0.5 mg/L thiamine, 100 mM aceto-syringone, and 3.5 g/L Phytigel, pH 5.2) and incubated in the dark at 21 °C for 3 d. After cocultivation, calli were transferred to resting medium (N6 salts and vitamins, 30 g/L maltose, 0.1 g/L myo-inositol, 0.3 g/L casein enzymatic hydrolysate, 0.5 g/L L-proline, 0.5 g/L L-glutamine, 2 mg/L 2,4-D, 0.5 mg/L thiamine, 100 mg/L timentin, and 3.5 g/L

Phytigel, pH 5.8) and incubated in the dark at 28 °C for 7 d. Calli were then transferred to selection medium (CIM plus 250 mg/L cefotaxime and 50 mg/L hygromycin B) and allowed to proliferate in the dark at 28 °C for 14 d. Well-proliferating tissues were transferred to CIM containing 75 mg/L hygromycin B. The remaining tissues were subcultured at 3- to 4-wk intervals on fresh selection medium. When a sufficient amount (about 1.5 cm in diameter) of the putatively transformed tissues was obtained, they were transferred to regeneration medium (MS salts and vitamins (Murashige and Skoog 1962), 30 g/L sucrose, 30 g/L sorbitol, 0.5 mg/L NAA, 1 mg/L BAP, and 150 mg/L cefotaxime) containing 40 mg/L hygromycin B and incubated at 26 °C, 16-h light, 90 μmol photons/m²/s. When regenerated plantlets reached at least 1 cm in height, they were transferred to rooting medium (MS salts and vitamins, 20 g/L sucrose, 1 g/L myo-inositol, and 150 mg/L cefotaxime) containing 20 mg/L hygromycin B and incubated at 26 °C under conditions of 16-h light (150 μmol photons/m²/s) and 8-h dark until roots were established and leaves touched the Phytatray II lid (Sigma-Aldrich, St. Louis, MO, USA). Plantlets were then transferred to soil.

Biolistic-mediated transformation

Embryogenic callus tissue pieces (3 to 4 mm) were transferred for osmotic pretreatment to CIM medium containing mannitol and sorbitol (0.2 M each). Four hours after treatment with osmoticum, tissues were bombarded as previously described (Cho et al. 2004) with modifications. Two milligrams of gold particles (0.6 μm), coated with 5 mg of plasmid DNA for editing rice *STOMAGEN*, was divided equally among 10 macrocarriers and used for bombardment with a Bio-Rad PDS-1000 He biolistic device (Bio-Rad, Hercules, CA, USA) at 650 psi. Sixteen to 18 h after bombardment, tissues were placed on osmotic-free CIM and incubated at 28 °C under dim light (10 to 30 μmol photons/m²/s, 16-h light). After 7 d, tissues were transferred to a selection medium (CIM containing 50 mg/L hygromycin B) and grown using the same procedure as described above, without timentin or cefotaxime supplemented in the media.

Validation of edits

T₀ plants targeted for edits in OsEPFL10 and OsSTOMAGEN were evaluated using PCR to amplify the region of interest. PCR products were Sanger sequenced. Sequence data were analyzed using the Synthego ICE tool (<https://ice.synthego.com/#/>) to detect alleles present (Conant et al. 2022). Only lines with homozygous frameshift mutations were retained for downstream experiments. Azygous T₂ plants were used for experimental data collection to minimize somaclonal variation, which may have accumulated during tissue culture (Bairu et al. 2010; Wei et al. 2016).

Phenotyping stomatal density, size, and aperture

Stomatal densities were recorded from epidermal impressions of leaves using nail polish peels (Kusumi et al. 2017).

Stomatal densities of 8 biological replicates of each leaf were taken from the widest section of fully expanded leaves. Images were taken using a Leica DM5000 B epifluorescent microscope at 10× magnification. Three images were collected per stomatal impression, and density per image was averaged. The number of stomata in a single stomatal band was counted, and the area of each band was measured (Huang et al. 2019). Stomatal densities were calculated by dividing stomatal counts by stomatal band area (mm^{-2}). Stomatal densities of the fifth leaf abaxial tissues were assayed for each allele of *epfl10* and *stomagen*. All subsequent phenotyping was conducted on plants of pooled alleles.

Epidermal peels of 21-d-old plants were produced using a razor blade on the adaxial leaf to remove tissues above the abaxial epidermal layer. Images of individual stomata at 100× magnification were captured. Guard cell length was measured using ImageJ. Thirty-five individual stomata from 5 biological replicates of each genotype were measured.

Stomatal aperture measurements were generated using epidermal peels of flag leaves from 85-d-old plants. Leaves were harvested at 1:00 PM and peels were generated immediately. Epidermal peels were then fixed by submerging in 4% (v/v) formaldehyde for 30 s using a method adapted by Eisele et al. (2016). Images of 20 individual stomata from 6 biological replicates of each genotype were measured.

Confocal microscopy images were captured using a Zeiss LSM 710 on epidermal peels of each genotype stained with propidium iodide for 40 s and immediately washed in water. Images were captured using 543 nm excitation laser and acousto-optic tunable filter of 5%. A 458/543 dichroic mirror was used to capture emission light. Photomultiplier tubes were allowed to collect wavelengths between 566 and 718 nm with a pinhole setting of 1 Airy unit. A 20× (0.8NA) objective was used. The primary gain was set to 748 with a digital gain of 2.5×. Images were processed using Zen 2.3 and Bitplane's Imaris to render Z images into a single plane.

Quantifying OsSTOMAGEN and OsEPFL10 transcript abundance

Total RNA was extracted from seedlings with the Qiagen Total RNAeasy Plant Kit at 3 developmental stages: 8 d after germination, 15 d after germination from the basal 2.5 cm above the leaf sheath of the fifth expanding leaf, and from the fully expanded leaf of 21-d-old leaves. RNA quality was validated on an agarose gel prior to reverse transcription using the QuantiTect reverse transcription kit to generate first-strand cDNA. Reverse transcription quantitative PCR (RT-qPCR) was performed using FAST SYBR on Applied Biosystem's QuantStudio 3 thermocycler. Relative expression levels were calculated by normalizing to the rice *UBQ5* housekeeping gene (LOC_Os01g22490) (Jain et al. 2018). Primers used for RT-qPCR are listed in Supplemental Table S2. Relative log fold expression was calculated using the $2^{-\Delta\Delta CT}$ method using OsSTOMAGEN in adult leaves as the control group. Three biological replicates were used per

tissue type. An unpaired *t* test was used to compare gene expression differences in each tissue type.

Determining methylation profile of genes of interest

Methylation profiles of rice genes of interest were viewed using the Plant Methylation Database (<https://epigenome.genetics.uga.edu/PlantMethylome/>) (Stroud et al. 2013; Niederhuth et al. 2016). Snapshots of CHH and CHG methylation 1.5 kb upstream of the start codon and 1.5 downstream of the stop codon were taken. The methylation data were collected on 3-wk-old rice leaf tissues.

Evolutionary analysis of STOMAGEN paralogs

Single-copy orthologs were searched using BUSCO v4.0.6 (Simão et al. 2015) and the viridiplantae_odb10 database for the species used in this study (Supplemental Table S1). Eighty-two orthologous groups present in at least 23 species were individually aligned with MAFFT v7.487 (–maxiterate 1000 –globalpair) (Katoh and Standley 2013). All multiple sequence alignments were concatenated, trimmed with TrimAl v1.4.rev15 (–gt 0.2) (Capella-Gutiérrez et al. 2009) and then used to infer a species tree with FastTree v2.1.10 (Price et al. 2010). The copy number variations of the STOMAGEN family were determined by searching for the stomagen domain (PF16851) from the protein annotation sets with hmmsearch v3.3 (Eddy 2011; Mistry et al. 2021) or from the genomes with exonerate v2.2.0 (Slater and Birney 2005) if genome annotations are absent. To understand the sequence variations of STOMAGEN and EPFL10 orthologs at the species and family level, we collected nonredundant *Oryza* or Poaceae species that have single copies of STOMAGEN or EPFL10 (Supplemental Table S2). The stomagen domain of STOMAGEN or EPFL10 orthologs was aligned with MAFFT, and the filtered alignment was used to compute normalized Shannon's entropy. Gaps were ignored.

Photosynthesis and stomatal conductance assays

Light response curves were generated using a LI6800 infrared gas analyzer (LI-COR, Lincoln, NE, USA) with chamber conditions set to the following: leaf temperature 25 °C; flow rate 500 $\mu\text{mol/s}$; water vapor pressure deficit 1.8 kPa; and CO_2 concentration of sample 400 $\mu\text{mol/mol}$. Light intensity was first increased to 2,000 $\mu\text{mol photons/m}^2/\text{s}$ and, with steady-state waiting times of 5 to 10 min, subsequently decreased to 1,500, 1,200, 1,000, 750, 500, 300, 200, 100, and 50 $\mu\text{mol photons/m}^2/\text{s}$ light. Light was composed of at least 90% red light and at maximum 40 $\mu\text{mol photons/m}^2/\text{s}$ blue light to match equipment specifications. Measurements were taken on fully expanded fifth leaves of 32-d-old plants grown in the greenhouse. Anatomical g_{max} was calculated using the double end-corrected version of the Franks and Farquhar (2001) equation from Dow et al. (2014) as described in Caine et al. (2019).

Physiological assays in Fig. 3, A and C were conducted on fully expanded leaf 5 of 21-d-old plants. Stomatal conductance and CO_2 assimilation data for Fig. 3C were captured

using an infrared gas analyzer (LI6400XT, LI-COR, Lincoln, NE, USA) with chamber conditions set to the following: light intensity 1,000 $\mu\text{mol photons/m}^2/\text{s}$ (90% red light and 10% blue light); leaf temperature 27 °C; flow rate 500 $\mu\text{mol/s}$; relative humidity 40%; and CO_2 concentration of sample 400 $\mu\text{mol/mol}$.

Thermal imaging

Thermal images were captured using a FLIR E8-XT Infrared Camera (FLIR-DIRECT, Wilmington, NC, USA). Images of well-watered and vegetative drought plants were taken 65 d after germination on the last day of the vegetative drought treatment. Images of reproductive drought plants were captured 102 d postgermination on the last day of the reproductive drought treatment. Images were captured between 1:00 PM and 2:00 PM to represent the effects of transpiration-mediated cooling during the hottest part of the day. Images were processed using FLIR Thermal Studio. Leaf temperatures of 4 to 6 leaves per biological replicate were quantified.

Water loss

Two plants of identical genotype were placed in a 10" $\times$ 10" flat and were covered with aluminum foil to minimize evaporation from soil. Daily water loss of each flat was calculated by taking the difference of flats with plants and the average of the flats without. Nonplant evaporation was estimated by measuring daily water loss from covered flats containing pots without plants. These control flats were distributed within the greenhouse assay area to reflect environmental variation. Eight replicate flats, with 2 plants per flat, for each genotype were measured daily from 70 to 77 d after germination. Pots remained well watered during the course of this experiment. Water loss was recorded, and flats were replenished, daily.

Graphs and statistics

All graphs were produced using the ggplot2 package in R studio (Wickham 2017). All statistics were calculated in R-studio using ANOVA and subsequent Tukey's honest significant difference (HSD) post hoc tests.

Accession numbers

Accession numbers for all genes and proteins considered in this study can be found in [Supplemental Table S1](#).

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Author contributions

N.G.K. developed the project idea and coordinated research efforts. N.G.K. constructed gene-editing vectors with assistance from D.D. M.T. and J.T. with oversight from M.J.C. produced *EPFL10* and *STOMAGEN* lines, respectively. K.S. generated dendrogram, gene tree, and all peptide models and designed all figures. K.S. with assistance from N.G.K. determined expression levels in carrot and sunflower. N.G.K. phenotyped stomatal densities, stomatal aperture, and stomatal size with assistance from A.G.C., S.L., and L.L. G.A.G. repeated stomatal density phenotyping independently to confirm results. N.G.K. phenotyped water loss and yield. N.G.K. and D.P.T. conducted g_s and A assays. D.P.T. captured thermal images. N.G.K. drafted the manuscript with edits from D.P.T., K.S., and K.K.N. K.K.N. provided LICOR equipment, thermal imaging equipment, and technical expertise. B.J.S. provided feedback on experimental design and the manuscript as well as the facilities to carry out experiments.

Supplemental data

The following materials are available in the online version of this article.

Supplemental Figure S1. Leaf 5 stomatal density of each *stomagen* and *epfl10* allele.

Supplemental Figure S2. Leaf 5 stomatal density of *stomagen epfl10* double knockout.

Supplemental Figure S3. Growth chamber stomatal conductance measurements at 1,000 $\mu\text{mol photons/m}^2/\text{s}$.

Supplemental Figure S4. Growth chamber carbon assimilation measurements at 1,000 $\mu\text{mol photons/m}^2/\text{s}$.

Supplemental Figure S5. Greenhouse stomatal conductance measurements at 1,000 $\mu\text{mol photons/m}^2/\text{s}$ under 2 watering regimes.

Supplemental Figure S6. Greenhouse carbon assimilation measurements at 1,000 $\mu\text{mol photons/m}^2/\text{s}$ under 2 watering regimes.

Supplemental Figure S7. OsEPFL10 editing vector map.

Supplemental Figure S8. OsSTOMAGEN editing vector map.

Supplemental Figure S9. Growth chamber iWUE efficiency measurements at 1,000 $\mu\text{mol photons/m}^2/\text{s}$.

Supplemental Table S1. List of the species used for comparative analyses and copy number variations of the STOMAGEN family.

Supplemental Table S2. List of primers used.

Supplemental File S1. OsEPFL10 editing vector sequence.

Supplemental File S2. OsSTOMAGEN editing vector.

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Conflict of interest statement. None declared.

Data availability

The data that support the findings of this study are available from the corresponding author, NGK, upon reasonable request.

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